

Total Questions: 35

LEVEL – II

Time: 1.5 hr.
Total Marks: 70

SECTION - 1

- This section contains **FIFTEEN (15)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 2

- This section contains **EIGHT (08)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each correct answer you will get **TWO** marks, if all the options are correct. If the correct answers chosen are less than the actual correct answers, partial marking (basis on weightage) will be given.
- There are **NO NEGATIVE** marks for wrong answers.

SECTION - 3

- This section contains **NINE (09)** questions of **PASSAGE TYPE OR COMPREHENSION TYPE**.
- Under each paragraph there will be three questions that has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 4

- This section contains **THREE (03)** questions. These questions are **MATCHING TYPE BASED QUESTIONS**.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers

Section - 1

1. Two students argue:

Student 1: "Light bends because its speed changes."

Student 2: "direction changes because Light bends."

Who is correct?

- A) Only 1
- B) Only 2
- C) Both are correct and related
- D) Both wrong

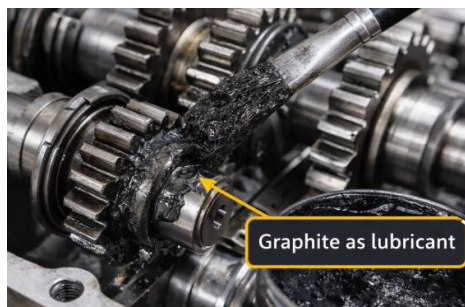
2. A materials scientist needs two ionic compounds:

- P → used for making heat-resistant ceramic
- Q → should have easy solubility in water

Which of the following pairs correctly satisfy the requirement based on lattice energy?

- A) P – MgO, Q – KBr
- B) P – NaCl, Q – MgO
- C) P – CaCl₂, Q – NaCl
- D) P – KBr, Q – MgO

3. Which property of graphite makes it suitable for this application?



- A) Strong covalent bonds between layers
 B) Weak forces between layers allowing them to slide
 C) Presence of four covalent bonds per carbon atom
 D) Ionic bonding between carbon atoms
4. Assertion (A): Increasing the refractive index of a prism increases both deviation and dispersion.
 Reason (R): All wavelengths experience equal increase in refractive index
- A) Both A and R are true, and R is correct explanation of A
 B) Both A and R are true, but R is NOT correct explanation
 C) A is true, R is false
 D) A is false, R is true
5. Consider the structure of the human heart. If a congenital defect allows a significant amount of blood to leak from the Left Ventricle directly into the Right Ventricle through a hole in the septum, what would be the most immediate physiological consequence for the patient?
- A) The lungs would receive highly oxygenated blood, leading to a massive increase in the rate of aerobic respiration.

- B) The systemic circulation (body organs) would receive less oxygenated blood, causing fatigue and shortness of breath.
 C) The blood pressure in the Aorta would increase significantly, leading to a higher risk of stroke.
 D) Deoxygenated blood would flow into the Left Atrium, preventing the bicuspid valve from closing

6. A person accidentally touches a hot iron. Before they even perceive the sensation of "pain" in their conscious mind, their hand has already jerked away. Which of the following sequences correctly describes the flow of the electrochemical signal through the nervous system for this specific action?
- A) Receptor → Motor Neuron → Spinal Cord → Sensory Neuron → Effector.
 B) Receptor → Sensory Neuron → Brain → Relay Neuron → Motor Neuron → Effector.
 C) Receptor → Sensory Neuron → Relay Neuron in Spinal Cord → Motor Neuron → Muscle.
 D) Effector → Relay Neuron → Sensory Neuron → Brain → Motor Neuron.
7. A group of students placed a germinating seed in a dark box with a small hole on one side through which light enters. After 4 days, the shoot bent toward the light, but the root grew vertically downward away from the light. Which statement correctly identifies the types of movements shown?
- A) Shoot: Positive Phototropism; Root: Negative Phototropism.

- B) Shoot: Negative Geotropism; Root: Positive Phototropism.
 C) Shoot: Positive Phototropism; Root: Positive Hydrotropism.
 D) Shoot: Nastic movement; Root: Tropic movement.

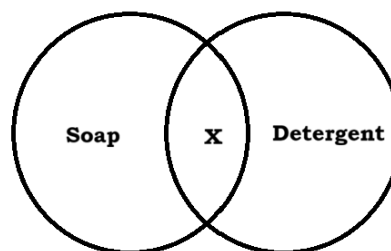
8. Which of the following statements accurately describes the mechanism of "Transpiration Pull" and its dependence on the diameter of Xylem vessels?

- A) Transpiration pull is a "pushing" force created by the roots that forces water up through large-diameter vessels.
 B) The pull is generated by the active contraction of Xylem Parenchyma cells surrounding the vessels.
 C) It is a seasonal process that only occurs when the atmospheric humidity is 100%.
 D) It is a suction force created by evaporation from leaves; it works most efficiently in narrow vessels due to the combined effects of cohesion and adhesion.

9. A convex lens forms a real image of an object placed at distance u . Without altering the lens or adding any optical device, which of the following change will necessarily make the image virtual?

- A) Decrease object distance such that $|u| < f$
 B) Increase refractive index of surrounding medium
 C) Decrease aperture of the lens
 D) Increase intensity of incident light

10. Observe the below Venn diagram.



Identify the correct placement of the following statements in the above Venn diagram.

- P. Used as cleansing agents
 Q. Ineffective in hard water
 R. Biodegradable
 S. Contain sodium salts with long hydrocarbon chains

	Soap	X	Detergent
A)	Q, R	P	S
B)	Q	P, S	R
C)	Q, R, S	P	-
D)	Q	P, S, R	-

11. A long straight wire carries 20 A current. A rectangular loop (length = 0.1 m, width = 0.05 m) carrying 5 A current is placed with its nearer side 2 cm from the wire. Find the net force on the loop.

- A) $0.714 \times 10^{-4} \text{ N}$
 B) $1.514 \times 10^{-4} \text{ N}$
 C) $2 \times 10^{-4} \text{ N}$
 D) $3 \times 10^{-4} \text{ N}$

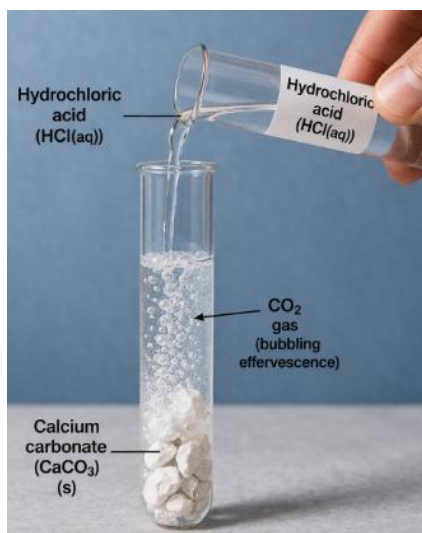
12. In a flowering plant, a pollen grain lands on a compatible stigma. If the distance between the stigma and the ovary is 15 mm, and the pollen tube grows at a constant rate of 1.5 mm/hour, how long will it take for the male germ cell to reach the female gamete?

- A) 1 hour
 B) 10 hours
 C) 15 hours
 D) 22.5 hours

13. Consider the process of "Triple Fusion" in flowering plants. Which cells are involved in this specific fusion, and what is the resulting product's primary function?

- A) One male gamete and one female egg; it forms the Zygote which becomes the embryo.
- B) The pollen tube and the stigma; it forms the nectar for insects.
- C) Two male gametes and one synergid; it forms the fruit wall.
- D) One male gamete and two polar nuclei; it forms the Endosperm which provides nutrition to the developing embryo.

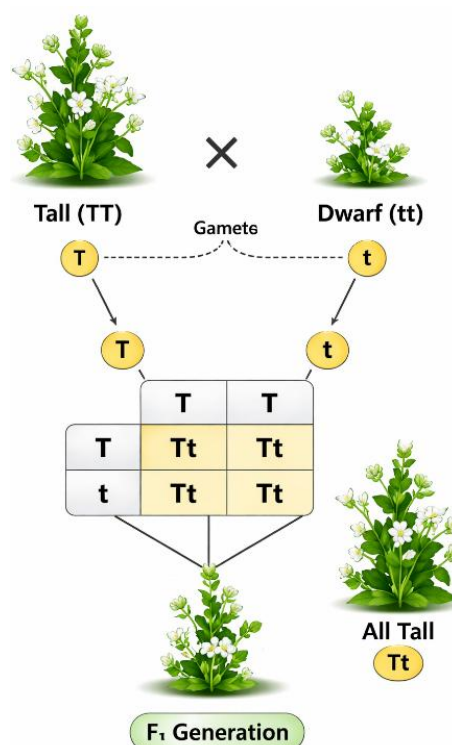
14. A student performs an experiment by reacting 15 g of calcium carbonate (CaCO_3) with 15 g of hydrochloric acid (HCl). Identify the limiting reagent, and the amounts of CO_2 and CaCl_2 formed.



- A) Limiting reagent: CaCO_3 ; $\text{CO}_2 = 6.6 \text{ g}$; $\text{CaCl}_2 = 16.65 \text{ g}$
- B) Limiting reagent: HCl ; $\text{CO}_2 = 9.04 \text{ g}$; $\text{CaCl}_2 = 20.8 \text{ g}$
- C) Limiting reagent: HCl ; $\text{CO}_2 = 6.6 \text{ g}$; $\text{CaCl}_2 = 16.65 \text{ g}$

D) Limiting reagent: CaCO_3 ; $\text{CO}_2 = 9.04 \text{ g}$; $\text{CaCl}_2 = 20.8 \text{ g}$

15. If the F_1 generation is self-pollinated to produce 800 offspring in the F_2 generation, how many of these offspring are expected to possess the same "Genotype" as the F_1 parent, and what is the biological reason for this?



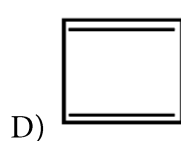
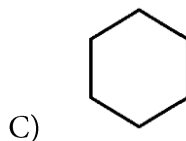
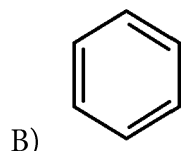
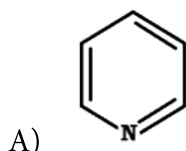
- A) 200 offspring; because the recessive trait disappears completely in the F_2 generation.
- B) 400 offspring; because the F_1 genotype is heterozygous (Tt), and the F_2 ratio for heterozygotes is 50%.
- C) 600 offspring; because the dominant phenotype (Tall) covers 75% of the total population.
- D) 800 offspring; because all offspring in the F_2 generation are genetically identical to the F_1 hybrid.

Section - 2

16. A resistor is connected to a battery for time t . Then resistance is halved and it is connected again for time t (same voltage). Which are correct?

- A) Heat produced in second case is double
- B) Total heat becomes 2 times initial heat
- C) Power becomes double
- D) Current becomes double

17. Which of the following compounds are aromatic?



18. A researcher is studying a specific autosomal dominant trait in a family. The father is heterozygous for the trait (Aa), and the mother is homozygous recessive (aa). Which of the following statements regarding their potential offspring are CORRECT?

- (A) There is a 50% probability that any given child will express the dominant phenotype.
- (B) All children will be carriers of the recessive allele, regardless of their phenotype.
- (C) It is biologically impossible for any of their children to be homozygous dominant (AA).
- (D) If they have four children, exactly two must be affected by the trait due to the 1:1 ratio.

19. A solenoid has 800 turns, length = 0.4 m, area = $2 \times 10^{-4} \text{ m}^2$. Current decreases from 5 A to 1 A in 0.01 s. Which are correct? (APPROX)

- (A) Magnetic field initially $\approx 5 \times 10^{-3} \text{ T}$
- (B) Change in flux $\approx 8 \times 10^{-7} \text{ Wb}$
- (C) Induced emf $\approx 0.064 \text{ V}$
- (D) If time doubles, emf halves

20. Consider the following statements.

I. Phenolphthalein turns pink in lime water.

II. Methyl orange turns yellow in acetic acid solution.

III. Blue litmus turns red in citric acid solution.

IV. Turmeric indicator remains yellow in magnesium hydroxide solution.

Which of the following options correctly identifies the true (T) and false (F) statements?

- A) II and IV are only false
- B) II is false
- C) I and III are only true
- D) I, III and IV are only true

21. In the human Endocrine System, many hormones work in "Antagonistic Pairs" to maintain Homeostasis (internal balance). Which of the following pairs and their functions are CORRECT?

- A) Insulin and Glucagon: Insulin lowers blood glucose, while Glucagon raises it by breaking down glycogen.
- B) Calcitonin and Parathyroid Hormone: They work together to regulate calcium levels in the blood and bones.

- C) Adrenaline and Noradrenaline: One prepares the body for stress, while the other puts the body to sleep immediately.
- D) Estrogen and Progesterone: They coordinate the menstrual cycle and pregnancy in females.

22. In the human "Female Reproductive System", the process of pregnancy involves a complex coordination of organs. Which of the following statements regarding fertilization and implantation are CORRECT?

- A) Fertilization typically occurs in the Fallopian tube (Oviduct) after the egg is released from the ovary.
- B) The zygote begins to divide rapidly as it moves down toward the uterus, forming a ball of cells.
- C) Implantation occurs when the embryo attaches itself to the thick, spongy lining of the uterus.
- D) The placenta is a specialized tissue that allows the maternal blood and fetal blood to mix directly to exchange gases.

23. In the human digestive system, the "Small Intestine" is considered the primary site for complete digestion and absorption. Which of the following structural or functional features support this?

- A) It receives bile from the liver, which contains enzymes that directly break down proteins into amino acids.
- B) The presence of millions of finger-like "villi" and "microvilli" drastically increases the surface area for nutrient absorption.
- C) It has an alkaline environment ($pH > 7$) created by bicarbonate ions in the

pancreatic juice, which is necessary for the activity of lipase.

D) The muscular walls perform peristalsis to move the chyme slowly, allowing sufficient time for enzyme action.

Section - 3

COMPREHENSION-1

In a specialized botanical garden, a researcher is studying a rare variety of the *Mirabilis* plant. Most plants have standard Green (G) leaves, but some exhibit a pale, silver-white phenotype called "Ghost" (g) foliage.

The researcher performs a series of controlled crosses:

- Cross 1: A pure-breeding Green plant is crossed with a pure-breeding Ghost plant. All 150 offspring in the F1 generation are Green.
- Cross 2: Two F1 Green plants are cross-pollinated. The resulting F2 generation consists of 442 Green plants and 148 Ghost plants.
- Cross 3: An F1 Green plant is crossed with the original parental Ghost plant.

24. Based on the data from Cross 1 and Cross 2, which of the following describes the nature of the "Ghost" foliage trait?

- A) It is a dominant trait because it reappears in the F2 generation.
- B) It is a recessive trait because it is masked in the F1 generation and follows a roughly 3:1 phenotypic ratio in the F2.

C) It is a co-dominant trait because the silver-white color is a blend of green and white pigments.

D) It is a lethal mutation that prevents 25% of the seeds from germinating at all.

25. In Cross 3 (F1 Green Parental Ghost), what is the expected phenotypic ratio of the offspring, and what is this type of cross specifically called?

A) 3 Green: 1 Ghost; Monohybrid F2 Cross.

B) 1 Green: 1 Ghost; Test Cross.

C) All Green; Back Cross to Dominant.

D) 1 Green: 2 Ghost; Incomplete Dominance Cross.

26. 3. If the researcher takes one of the 148 "Ghost" plants from the F2 generation and allows it to self-pollinate for three more generations, what will be the leaf color of the resulting progeny?

A) They will eventually revert to 100% Green as the plant matures.

B) They will show a 3:1 ratio of Green to Ghost in every generation.

C) They will be 100% Ghost-foliaged because the plant is homozygous recessive (gg) and breeds true.

D) They will all die because recessive traits cannot survive multiple rounds of self-pollination.

COMPREHENSION-2

Metals are extracted based on their position in the activity series using methods such as heating, displacement, electrolysis, or reduction of their oxides with carbon or carbon monoxide after roasting.

27. A metallurgical plant needs to extract sodium metal efficiently. Which of the following methods should be used?

A) Heating sodium oxide with carbon

B) Electrolysis of molten sodium chloride

C) Displacement using zinc

D) Roasting followed by reduction

28. An ore of zinc is first converted into oxide and then heated with carbon to obtain the metal. Which of the following statements is correct?

A) Zinc lies in the middle of the activity series

B) Carbon acts as a reducing agent

C) Electrolysis is the preferred method

D) Roasting is involved before reduction

29. A jeweller obtains silver from its ore by heating and sometimes by displacement reactions. Which of the following inferences are correct?

(i) Silver is highly reactive

(ii) Silver lies at the bottom of the activity series

(iii) Silver can be found in native state

(iv) Electrolysis is necessary for extraction

A) (i) and (ii) only B) (ii) and (iii) only

C) (iii) and (iv) only D) (i) and (iv) only

COMPREHENSION-3

A precision compound microscope is adjusted so that the final image is formed at infinity, allowing relaxed viewing. The objective has a focal length of 1 cm and the eyepiece has a focal length of 4 cm. The distance between the two lenses is 17 cm. The object is placed very close to the focal point of the objective, producing a highly enlarged real intermediate image that lies exactly at the focal plane of the eyepiece.

During observation, a slight mechanical disturbance causes the intermediate image to shift 0.5 cm closer to the eyepiece. The observer does not readjust the eyepiece and continues viewing under the same conditions.

Based on this situation, answer the following:

30. After the disturbance, the student notices that the image is no longer clearly visible with a relaxed eye. Which of the following correctly explains the new optical condition?

- A) The final image is no longer formed at finite and shifts to infinity distance
- B) The eye must now accommodate to bring the image into focus
- C) The angular magnification increases due to reduced image distance
- D) The final image distance becomes less than zero

31. Before the disturbance, the microscope was in normal adjustment. Considering practical optical constraints and effective tube length, the magnification of the microscope is closest to:

- A) 68.45
- B) 62.34
- C) 81.25
- D) 96.34

32. The primary reason for the observed image blur after the disturbance is best explained by:

- A) A change in focal lengths due to misalignment
- B) The intermediate image no longer lying at the focal plane of the eyepiece
- C) Failure of the objective to produce a real image
- D) Change in nature of image formed by eyepiece

Section - 4

33. Match Column A with Column B

Column A	Column B
(A) Resistance of a conductor	(P) Doubles according to Ohm's law
(B) Resistivity of material	(Q) Depends on length and area of cross-section
(C) Electrical power (in terms of V and R)	(R) Given by V^2/R
(D) Commercial unit of energy	(S) kWh, used in electricity billing
(E) Current in a conductor when voltage is doubled (R constant)	(T) Independent of dimensions but depends on nature of material

A) A-Q, B-T, C-R, D-S, E-P

B) A-T, B-Q, C-R, D-P, E-S

C) A-Q, B-T, C-P, D-S, E-R

D) A-R, B-Q, C-T, D-S, E-P

34. Match Column A with Column B

Type of Reaction	Example
(A) Exothermic reaction	(P) Formation of curd from milk
(B) Precipitation reaction	(R) $4\text{Fe} + 3\text{O}_2 + 6\text{H}_2\text{O} \rightarrow 2\text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$
(C) Neutralisation reaction	(Q) $\text{C}_{25}\text{H}_{52} + 38\text{O}_2 \rightarrow 25\text{CO}_2 + 26\text{H}_2\text{O} + \text{Energy}$
(D) Redox reaction	(S) $\text{Mg}(\text{OH})_2 + 2\text{HCl} \rightarrow \text{MgCl}_2 + 2\text{H}_2\text{O}$

- A) A – P, B – Q, C – R, D – S
 B) A – Q, B – P, C – S, D – R
 C) A – R, B – S, C – Q, D – P
 D) A – Q, B – S, C – R, D – P

35. Match Column A with Column B

Column I (Plant Hormone)	Column II (Physiological Effect)
(A) Auxin	(P) Breaking seed dormancy and bolting (stem elongation)
(B) Gibberellin	(Q) Promoting cell division and delaying leaf senescence
(C) Cytokinin	(R) Curving of shoots toward light (Phototropism)
(D) Ethylene	(S) Acceleration of fruit ripening and leaf abscission

- A) A-R, B-Q, C-P, D-S
 B) A-P, B-R, C-Q, D-S
 C) A-R, B-P, C-Q, D-S
 D) A-S, B-P, C-Q, D-R